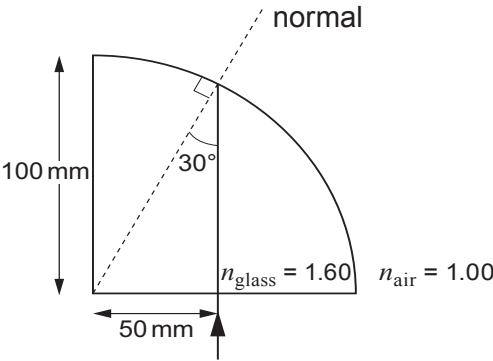


Examiner  
only

7. (a) (i) A narrow beam of light enters a glass block of quarter-circle cross-section, as shown. The refractive index of the glass is 1.60.



Calculate the angle of refraction into the air at the curved surface and carefully sketch the refracted beam **on the diagram**. [Use of a protractor is not required.][3]

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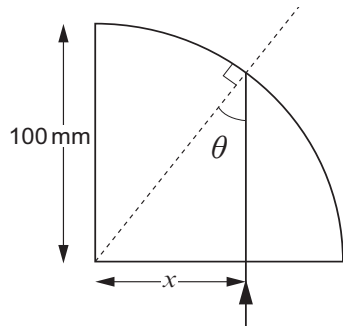
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- (ii) Calculate the largest distance,  $x$ , along the bottom face of the block, at which the beam can enter the block normally, for it to emerge from the curved face. You should refer to the angle  $\theta$  in your calculation. [3]



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Examiner  
only

(b) Explain how a multimode (thick) glass fibre transmits light, and why rapid streams of data cannot be transmitted successfully through long lengths of multimode fibre. [6 QER]

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